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Sri Sivasubramaniya Nadar College of Engineering, Kalavakkam – 603 110.

(An Autonomous Institution, Affiliated to Anna University, Chennai)

B.E. / B.Tech. End Semester Theory Examinations, April / May 2024.

Sixth Semester

Computer Science and Engineering

UCS2604 PRINCIPLES OF MACHINE LEARNING

(Regulations 2021)

Time: **Three Hours**Maximum: **100 Marks**

K1: Remembering

K2: Understanding

K3: Applying

K4: Analyzing

K5: Evaluating

CO1:	Explain the basic concepts of machine learning
CO2:	Apply supervised algorithms for different classification problems
CO3:	Explain the need for ensemble methods
CO4:	Apply unsupervised and reinforcement learning techniques to various problems
CO5:	Apply dimensionality reduction and optimization techniques

Part – A (5 × 2 = 10 Marks)

		KL	CO	PI
1.	State the use of confusion matrix.	K1	CO1	1.4.1
2.	Define support vectors.	K1	CO2	1.4.1
3.	What does k signify in the K-NN algorithm?	K2	CO3	1.4.1
4.	Define Markov Decision Process.	K2	CO4	1.4.1
5.	PCA is an unsupervised learning approach. Justify.	K1	CO5	1.4.1

Part – B (5 × 6 = 30 Marks)

		KL	CO	PI
6.	Describe the method for creating training and test data. Summarize the effect of bias and variance in training and testing.	K2	CO1	1.1.1 1.4.1 2.1.3
7.	With a diagram, describe how McCulloch and Pitts neuron work.	K3	CO2	2.1.3 2.3.1
8.	Explain overfitting problem of the decision tree.	K3	CO3	1.4.1
9.	Outline the steps of Q-Learning.	K2	CO4	2.3.1 2.4.1
10.	Compare PCA and LDA in terms of their similarities and dissimilarities.	K2	CO5	1.4.1

Part – C (5 × 12 = 60 Marks)

		KL	CO	PI																																												
11.	The following table gives a dataset about stolen vehicles. Using Naive Bayes classifier, classify the new data {Color: Red, Type: SUV, Origin: Domestic} <table border="1"> <thead> <tr> <th>Color</th> <th>Type</th> <th>Origin</th> <th>Stolen</th> </tr> </thead> <tbody> <tr><td>Red</td><td>Sports</td><td>Domestic</td><td>Yes</td></tr> <tr><td>Red</td><td>Sports</td><td>Domestic</td><td>No</td></tr> <tr><td>Red</td><td>Sports</td><td>Domestic</td><td>Yes</td></tr> <tr><td>Yellow</td><td>Sports</td><td>Domestic</td><td>No</td></tr> <tr><td>Yellow</td><td>Sports</td><td>Imported</td><td>Yes</td></tr> <tr><td>Yellow</td><td>SUV</td><td>Imported</td><td>No</td></tr> <tr><td>Yellow</td><td>SUV</td><td>Imported</td><td>Yes</td></tr> <tr><td>Yellow</td><td>SUV</td><td>Domestic</td><td>No</td></tr> <tr><td>Red</td><td>SUV</td><td>Imported</td><td>No</td></tr> <tr><td>Red</td><td>Sports</td><td>Imported</td><td>Yes</td></tr> </tbody> </table>	Color	Type	Origin	Stolen	Red	Sports	Domestic	Yes	Red	Sports	Domestic	No	Red	Sports	Domestic	Yes	Yellow	Sports	Domestic	No	Yellow	Sports	Imported	Yes	Yellow	SUV	Imported	No	Yellow	SUV	Imported	Yes	Yellow	SUV	Domestic	No	Red	SUV	Imported	No	Red	Sports	Imported	Yes	K3	CO1	1.1.1 2.1.3 13.3.1
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12.	For a classifier, the ground truth and the predicted output are given below Ground Truth 1 0 1 0 1 1 1 1 0 Trained classifier 0 0 1 0 0 1 0 1 1 Calculate the confusion matrix, accuracy, precision, and recall.	K3	CO1	1.1.1 2.1.3 13.3.1																																												
13.	Describe Forward and Backward propagation in Multilayered Perceptron. Justify the use of learning rate in perceptron networks.	K2	CO2	2.4.1 13.3.1																																												
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14.	Describe the concept and working of Support Vector Machine (SVM). How do we use SVM for multi-class classification?	K2	CO2	2.4.1 13.3.1																																												

15.	The table given below shows the movie ticket booking pattern. Construct the decision tree and find the root node of the tree. <table border="1"> <thead> <tr> <th>Show Time</th><th>Shot</th><th>Actor</th><th>Genre</th><th>Book</th></tr> </thead> <tbody> <tr><td>MATINEE</td><td>IN DOOR</td><td>ALLU ARJUN</td><td>DRAMA</td><td>No</td></tr> <tr><td>MATINEE</td><td>IN DOOR</td><td>MOHANLAL</td><td>ACTION</td><td>No</td></tr> <tr><td>EVENING</td><td>IN DOOR</td><td>RAJANIKANTH</td><td>DRAMA</td><td>No</td></tr> <tr><td>EVENING</td><td>IN DOOR</td><td>MOHANLAL</td><td>COMEDY</td><td>No</td></tr> <tr><td>EVENING</td><td>OUT DOOR</td><td>ALLU ARJUN</td><td>COMEDY</td><td>No</td></tr> <tr><td>EVENING</td><td>OUT DOOR</td><td>RAJANIKANTH</td><td>COMEDY</td><td>No</td></tr> <tr><td>EVENING</td><td>IN DOOR</td><td>ALLU ARJUN</td><td>ACTION</td><td>No</td></tr> <tr><td>EVENING</td><td>OUT DOOR</td><td>MOHANLAL</td><td>COMEDY</td><td>No</td></tr> <tr><td>MATINEE</td><td>IN DOOR</td><td>MOHANLAL</td><td>DRAMA</td><td>No</td></tr> <tr><td>MATINEE</td><td>OUT DOOR</td><td>RAJANIKANTH</td><td>COMEDY</td><td>Yes</td></tr> <tr><td>EVENING</td><td>OUT DOOR</td><td>MOHANLAL</td><td>ACTION</td><td>Yes</td></tr> <tr><td>MATINEE</td><td>OUT DOOR</td><td>ALLU ARJUN</td><td>ACTION</td><td>Yes</td></tr> <tr><td>MATINEE</td><td>IN DOOR</td><td>MOHANLAL</td><td>COMEDY</td><td>Yes</td></tr> <tr><td>EVENING</td><td>OUT DOOR</td><td>RAJANIKANTH</td><td>ACTION</td><td>Yes</td></tr> </tbody> </table>	Show Time	Shot	Actor	Genre	Book	MATINEE	IN DOOR	ALLU ARJUN	DRAMA	No	MATINEE	IN DOOR	MOHANLAL	ACTION	No	EVENING	IN DOOR	RAJANIKANTH	DRAMA	No	EVENING	IN DOOR	MOHANLAL	COMEDY	No	EVENING	OUT DOOR	ALLU ARJUN	COMEDY	No	EVENING	OUT DOOR	RAJANIKANTH	COMEDY	No	EVENING	IN DOOR	ALLU ARJUN	ACTION	No	EVENING	OUT DOOR	MOHANLAL	COMEDY	No	MATINEE	IN DOOR	MOHANLAL	DRAMA	No	MATINEE	OUT DOOR	RAJANIKANTH	COMEDY	Yes	EVENING	OUT DOOR	MOHANLAL	ACTION	Yes	MATINEE	OUT DOOR	ALLU ARJUN	ACTION	Yes	MATINEE	IN DOOR	MOHANLAL	COMEDY	Yes	EVENING	OUT DOOR	RAJANIKANTH	ACTION	Yes	K4	CO3	1.1.1 2.1.3 2.4.1 13.3.1
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16.	Construct a Kd-Tree with the following points: $A(2, 3)$, $B(4, 2)$, $C(4, 5)$, $D(3, 3)$, $E(1, 5)$, $F(4, 4)$. Further, show how the 2D plane is bisected as points are inserted. On the axes, draw each point as well as the horizontal or vertical lines that bisect the plane through each point. <i>Note: Be careful when inserting $(4, 4)$ as ties are treated the same as greater than.</i>	K4	CO3	1.1.1 2.1.3 2.4.1 13.3.1																																																																											
17.	Cluster the following eight points (with (x, y) representing locations) into three clusters using k-means clustering and find the three cluster centers after the second iteration. The points and their coordinates are $A1(2, 10)$, $A2(2, 5)$, $A3(8, 4)$, $A4(5, 8)$, $A5(7, 5)$, $A6(6, 4)$, $A7(1, 2)$, $A8(4, 9)$. Initial cluster centers are: $A1(2, 10)$, $A4(5, 8)$ and $A7(1, 2)$	K4	CO4	2.3.1 2.4.1																																																																											
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18.	Prove that Q-learning and SARSA solves the problem of cliff walking environment using optimal and safer paths.	K4	CO4	2.3.1 2.4.1																																																																											

19.	Compute the principal components and reduce the dimensions. <table border="1"> <thead> <tr> <th>Features</th><th>Sample 1</th><th>Sample 2</th><th>Sample 3</th><th>Sample 4</th></tr> </thead> <tbody> <tr> <td>x</td><td>4</td><td>8</td><td>13</td><td>7</td></tr> <tr> <td>y</td><td>11</td><td>4</td><td>5</td><td>14</td></tr> </tbody> </table>	Features	Sample 1	Sample 2	Sample 3	Sample 4	x	4	8	13	7	y	11	4	5	14	K3	CO5	1.1.1 2.1.3 2.4.1 13.3.1
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20.	Consider the given samples belonging to two classes, W1 and W2. Apply LDA to transform the data into k-dimensional space. $W1: \begin{pmatrix} 4 \\ 3 \end{pmatrix}, \begin{pmatrix} 2 \\ 5 \end{pmatrix}, \begin{pmatrix} 4 \\ 8 \end{pmatrix}$ $W2: \begin{pmatrix} 6 \\ 2 \end{pmatrix}, \begin{pmatrix} 11 \\ 3 \end{pmatrix}, \begin{pmatrix} 9 \\ 4 \end{pmatrix}$	K3	CO5	1.1.1 2.1.3 2.4.1 13.3.1															